Chapter 3

Push Buttons, Switches and Sensors

3.1 Introduction

Purpose of Push Buttons and Switches

- Provide **manual control** of electrical devices
- Most common form of **manual control**
- Control motors through **motor starters**
- Open or close contacts by pressing a button

3.2 Push Buttons

Definition

A device that provides control of a motor through a **motor starter** by pressing a button that opens or closes contacts.

Key Characteristic

- **Momentary contact devices** (most common)
- Make or break connection **only while pressure is applied**
- When pressure is removed, button returns to **normal position**

Contact Types

Type	Symbol	Description
Normally Open (NO)	—| |——	Movable contact above, not touching stationary contact
Normally Closed (NC)	—|\ |——	Movable contact below, touching stationary contact

3.2.1 Normally Open (NO) Push Button

Characteristics:

- Movable contact is drawn **above and not touching** the stationary contact
- Circuit is **open** in normal state
- No current flows when not pressed

Operation:

- **Press** → Movable contact bridges stationary contacts → **Circuit closes**
- **Release** → Spring returns contact → **Circuit opens**

3.2.2 Normally Closed (NC) Push Button

Characteristics:

- Movable contact is drawn **below and touching** the stationary contact
- Circuit is **closed** in normal state
- Current flows when not pressed

Operation:

- **Press** → Movable contact moves away → **Circuit opens**
- **Release** → Spring returns contact → **Circuit closes**

3.3 Types of Push Buttons

1. Double Acting Push Button

Description:

- Contains **both NO and NC contacts**
- Two movable contacts connected by **one common shaft**

Operation:

Action	Top Contact (NO)	Bottom Contact (NC)
Pressed	Closes	Opens
Released	Opens	Closes

> **Important:** Must connect wires to the correct set of contacts.

```
═══╗      ╔═══     (NO contact - opens when pressed)
    ║    ║
═══║    ║═══
    ║    ║
    ║    ║
═══║    ║═══     (NC contact - closes when pressed)
    ║    ║
═══╝      ╚═══
      ↓
 Common Shaft
```

Applications: Start/Stop control in single button · Motor control circuits

2. Stacked Push Button

Description:

- Contains **multiple contacts** controlled by a **single push button**
- Made by connecting multiple contact units together

Example – Emergency Stop: One stop button can **stop three motors** at the same time; each contact controls a different motor circuit.

```
     ┌─────┐
     │     │    Motor 1 Stop
─────┼─────┼────
     │     │
     │     │    Motor 2 Stop
─────┼─────┼────
     │     │
     │     │    Motor 3 Stop
─────┼─────┼────
     └──┬──┘
        │      Single Push Button
    ────┴────
```

Applications: Multiple circuit control · Emergency stop for multiple machines · Master control stations

3. Push-Pull Button

Description:

- Contains **both NO and NC contacts**
- Provides both **START and STOP** functions in **one button**
- Different contact arrangement from double acting

Operation:

Action	NC Contact	NO Contact
Pull	Remains closed	Closes (bridges contacts)
Release (after pull)	Remains closed	Returns open
Push	Opens	Remains open

Operation Sequence:

1. **Pull the button:** NO contact bridges stationary contacts → **Circuit completes** (START); NC contact remains closed → Holds circuit
2. **Push the button:** NC section opens → **Circuit breaks** (STOP); NO section remains open

> **Advantage:** Saves panel space by combining start/stop in one button.

4. Lighted (Illuminated) Push Button

Description: Mechanical switch with **built-in light source** (typically LED); indicates switch status; easy to find in dark environments.

Features:

- Indicates motor **running, stopped, or tripped on overload**
- Equipped with small transformer to reduce voltage
- **Lens caps** of different colors available
- Provides **momentary** and **latching** actions

Color Indications:

Color	Typical Meaning
Green	Run/Start/ON
Red	Stop/Off/Fault
Amber	Warning/Caution

White	General indication

3.4 Switches

Definition

An electrical device used to **make or break** an electrical circuit either **manually** or **automatically**.

Classification Based on Poles and Throws

Term	Meaning
Poles	Number of circuits controlled by the switch
Throws	Number of positions the switch can adopt

3.4.1 Types Based on Poles and Throws

1. SPST (Single Pole Single Throw)

Feature	Description
Poles	1
Throws	1
Action	ON or OFF

```
────────┐    ┌────────
────────┘    └────────
```

Application: Simple ON/OFF control

2. SPDT (Single Pole Double Throw)

Feature	Description
Poles	1
Throws	2
Action	Selects between two circuits

```
          ┌── Position 1
──────────┤
          └── Position 2
```

Application: Selector switches, changeover

3. DPST (Double Pole Single Throw)

Feature	Description
Poles	2
Throws	1
Action	Switches two circuits simultaneously

```
──────────┐     ┌──────────
──────────┤     ├──────────
          │     │
──────────┤     ├──────────
──────────┘     └──────────
```

Application: Double line switching, both ON/OFF together

4. DPDT (Double Pole Double Throw)

Feature	Description
Poles	2
Throws	2
Action	Two circuits with two positions each

```
──────────┐     ┌── Position 1
          │     ├── Position 2
──────────┤     │
          │     ├── Position 1
──────────┘     └── Position 2
```

Application: Motor forward/reverse

3.5 Types of Switches by Application

3.5.1 Pressure Switch

Definition: Switch **controlled by pressure**, used to **monitor and control fluid pressure**. **Mechanically operated**.

Applications: HVAC systems · Hydraulic and pneumatic systems · Water pumps · Air compressors · Industrial machinery

Contact Types:

Type	Description
NO	Closes when pressure is reached
NC	Opens when pressure is reached
CO (Changeover)	Both NO and NC contacts

Working Principle:

```
    [Fluid Pressure]
           ↓
    [Piston/Diaphragm]
           ↓
    [Spring Balance]
           ↓
    [Contact Operation]
```

1. Fluid pressure acts on one side of piston
2. Atmospheric pressure on opposite side
3. Preloaded spring balances fluid force
4. **Setpoint screw** adjusts activation pressure
5. When pressure reached → Piston moves → Contacts change

> **Adjustment:** Setpoint screw changes spring pre-compression for higher/lower activation pressure.

Two Main Types:

Type	Description
Mechanical	Uses mechanical components (spring, piston)
Electronic	Uses electronic sensors and circuits

3.5.2 Float Switch

Definition: Level switch (liquid level sensing), **mechanically operated**, used in **vessels** to sense liquid levels.

Applications: Open/close piping solenoid valves · Control pumps (AC and DC) · Pump motor magnetic starters · Transfer liquids from pumps to tanks

Working Principle:

1. Float is **weighted**
2. As liquid **rises**, switch floats and turns upside down
3. Internal contacts actuate
4. Controls AC/DC pump motor magnetic starters

```
    [Liquid Level Rising]
           ↓
    [Float Switches Position]
           ↓
    [Contacts Actuate]
           ↓
    [Control Action]
```

3.5.3 Limit Switch

Definition: Electromechanically operated switch, **actuated by physical contact** with an object. Wedge symbol represents the **bumper arm**.

Components: 1. **Actuator** – interacts with the object 2. **Switch body** – houses contacts

Applications: Control circuits of machine processes · Govern **starting, stopping, or reversal** of motors · Position sensing · End-of-travel detection

Types of Limit Switches:

Type	Description
Roller Lever	Uses roller for smooth actuation
Plunger	Linear motion actuation
Spring Arm	Flexible arm for various angles
Proximity	Non-contact sensing (electronic)

3.5.4 Flow Switch

Definition: Mechanically operated switch that **monitors fluid flow** in a system. Acts as a **protection device**.

Function: Warns of liquid flow problems · Detects **failing pump** · Detects **defective valve**

Type	Symbol/Name	Application
Liquid Flow Switch	—	Liquid flow monitoring
Air Flow Switch	Sail Switch	Air flow monitoring

Circuit Connections: Energize motor starter coils · Energize relays · Energize indicating lights

3.5.5 Temperature Switch (Thermostat)

Definition: Electronic device that **monitors and controls temperature**, also known as a **thermostat**. Contacts change state based on **temperature threshold**.

Applications: HVAC systems · Industrial process control · Automotive systems · Refrigeration

Function: Maintains desired temperature range; actuated by **environmental temperature change**; controls heating or cooling.

```
Temperature Rising Above Setpoint  → Contacts Open  → Cooling Turns ON
Temperature Falling Below Setpoint → Contacts Close → Heating Turns ON
```

3.5.6 Selector Switch

Definition: Manually operated switch, **rotated** (not pushed) to open/close contacts. Maintains contact positions.

> **Key Difference:** Different operator mechanism from push buttons.

Common Types:

Type	Positions
Two-position	ON/OFF, RUN/STOP
Three-position	START/OFF/STOP
Momentary	Spring return to center

Common Selector Switch: MAN–OFF–AUTO

- **Single-pole double-throw (SPDT)** with center OFF position
- Three positions: **MAN**, **OFF**, **AUTO**

Position	Function
MAN (Manual)	Manual control of process
OFF	Process stopped
AUTO	Automatic control

3.6 Comparison: Push Buttons vs Selector Switches

Feature	Push Button	Selector Switch
Operation	Pushed (in/out)	Rotated
Contact type	Momentary (usually)	Maintained (usually)
Return mechanism	Spring return	Detent/latching
Positions	2 (pressed/released)	Multiple (2, 3, or more)
Applications	Start/Stop commands	Mode selection

3.7 Safety Considerations

Push Button Safety

1. **Emergency Stop** – Must be easily accessible
2. **Color coding** – Red for stop/emergency
3. **Mushroom head** – Emergency stops
4. **Key lock** – Prevent unauthorized operation

Switch Safety

1. **Proper rating** – Must handle circuit current
2. **Environmental protection** – NEMA ratings
3. **Proper grounding** – For safety
4. **Guard barriers** – Prevent accidental actuation

3.8 Key Terminology

Term	Definition
Momentary	Contact only while actuated
Maintained	Contact stays in position until changed
Poles	Number of circuits controlled
Throws	Number of positions per pole
Actuator	Mechanism that operates the switch
Contact block	Assembly containing switch contacts
Lens cap	Colored cover for illuminated push buttons

Review Questions

1. What is the difference between a normally open and normally closed push button?
2. Draw and explain the operation of a double-acting push button.
3. What is the purpose of a stacked push button?
4. Explain the operation of a push-pull button.
5. Differentiate between SPST and DPDT switches.
6. What is the purpose of a pressure switch and how does it work?
7. Explain the working principle of a float switch.
8. What applications use limit switches?
9. What is the MAN-OFF-AUTO selector switch used for?
10. Why are lighted push buttons used in industrial applications?

END OF CHAPTER 3 NOTES